

Running and Deploying `sudoku_app.py`

Local Testing and Streamlit Community Cloud

Overview

This guide covers two things: running `sudoku_app.py` **locally** for quick testing while developing, and deploying it to the **web** via Streamlit Community Cloud, the standard free way to put a Streamlit app online with a shareable public URL. Test locally first; it catches most problems (missing files, import errors, a broken button).

1 Local Testing

Running the app locally uses the exact same code as the deployed version, so anything that works locally will work once deployed, and vice versa.

1.1 Required Files

All of the following must sit in the same folder:

- `sudoku_app.py` – the Streamlit application
- `sudoku_solver.py`
- `utils.py` and `logic.py`
- `puzzles.json`
- `requirements.txt`

1.2 Running the App

Perform the following steps. Do Steps 1 and 2 only once. Do Step 3 to start the app.

1. Create and activate a virtual environment:

```
python3 -m venv .venv
source .venv/bin/activate      # Windows: .venv\Scripts\activate
```

2. Install dependencies:

```
pip install -r requirements.txt
```

3. Run the app:

```
streamlit run sudoku_app.py
```

Streamlit opens the app at `http://localhost:8501` in your default browser. Edits to `sudoku_app.py` are picked up on save. Streamlit shows a “Rerun” prompt in the browser tab (or reruns automatically, depending on your Streamlit version’s settings) rather than requiring you to restart the server.

1.3 Stopping the App

Press `Ctrl+C` in the terminal running `streamlit run`. The virtual environment in `.venv` can be left in place; the next `./run_local.sh` (or manual `source .venv/bin/activate`) reuses it without reinstalling anything.

1.4 Common Local Issues

- **ModuleNotFoundError:** No module named ‘streamlit’ (or ‘networkx’). The virtual environment wasn’t activated, or `pip install -r requirements.txt` wasn’t run inside it. Re-run Option A, or repeat Option B’s steps 1–2.
- **FileNotFoundError for `puzzles.json`.** The app reads it via a relative path, so `streamlit run sudoku_app.py` must be run from the folder containing `puzzles.json` – not a parent or sibling folder.
- **Port already in use.** Another Streamlit app (or a previous run that didn’t shut down cleanly) is already using port 8501. Either stop it, or run `streamlit run sudoku_app.py --server.port 8502` to use a different port.

2 Prepare the Repository (for Web Deployment)

Create a public GitHub repository containing the following files, all in the same folder:

- `sudoku_app.py`
- `sudoku_solver.py`
- `utils.py` and `logic_.py`
- `puzzles.json`
- `requirements.txt`

Requirements

Contents of the `requirements.txt` file:

```
streamlit
networkx
```

`networkx` is required even though it is unused in `logic_.py`; it is imported at the top of that file, and Streamlit Cloud’s build step will install it via `pip` rather than needing a stub, as a local sandbox might.

3 Create a Streamlit Cloud Account

Go to <https://share.streamlit.io> and sign in with GitHub. A GitHub account is required either way, since deployment is GitHub-based.

4 Deploy the App

1. From the Streamlit Community Cloud page, choose **Deploy a public app from GitHub**.
2. Click **Deploy now**.
3. Fill in your repository, branch, and the entrypoint file path (`sudoku_app.py`) – or paste the GitHub URL to the file directly.
4. Optionally set a custom subdomain. If left blank, Streamlit Cloud assigns one automatically, e.g. `yourname-reponame-sudoku-app-xxxxx.streamlit.app`.
5. Click **Deploy**.

5 Watch It Build

Most apps are live within a few minutes. After the initial deployment, every push to the repository updates the live app automatically – no redeploy step is needed. Build logs are visible to the repository owner if anything fails, e.g. a missing dependency.

6 Share the URL

Paste the deployed Streamlit app URL into the designated section of `Sudoku_Assignment.ipynb`.

Notes Specific to This App

- `sudoku_app.py` reads `puzzles.json` via a relative path and imports `sudoku_solver.py`, so these files must sit in the repository root together (not in a subfolder), unless the paths in the code are updated accordingly.
- No secrets or environment variables are needed for this app, so the “Advanced settings” step during deployment can be skipped entirely.